

HTTPS Traffic Analysis Using tcpdump

Objective

The objective of this simulated home lab is to capture and analyze the network traffic generated while accessing an HTTPS website from an Ubuntu virtual machine using tcpdump. The lab focuses on identifying the TCP three-way handshake, encrypted HTTPS communication, multiple TCP connections, and the overall sequence of events involved in secure web communication.

Lab Setup

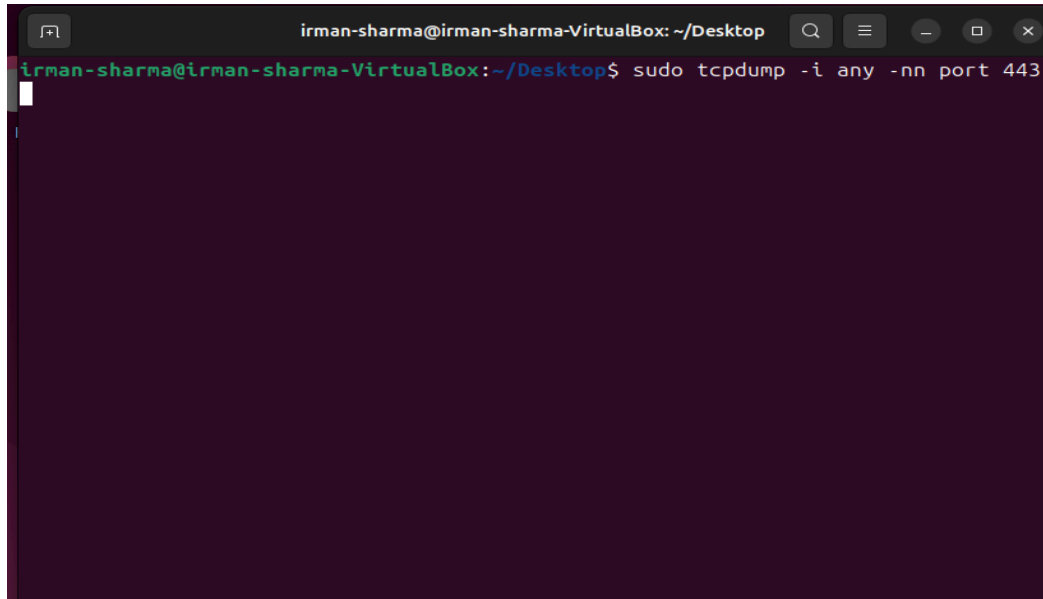
- Windows Host
- Ubuntu VM
- Firefox
- tcpdump
- VirtualBox NAT networking

Procedure

- Started the Ubuntu Virtual Machine
- Opened Firefox browser
- Opened the terminal, and ran tcpdump command to start packet capture
- Visited **https://example.com/**
- Allowed the page to load
- Stopped the capture with Ctrl + C
- Reviewed the packet capture

Analysis

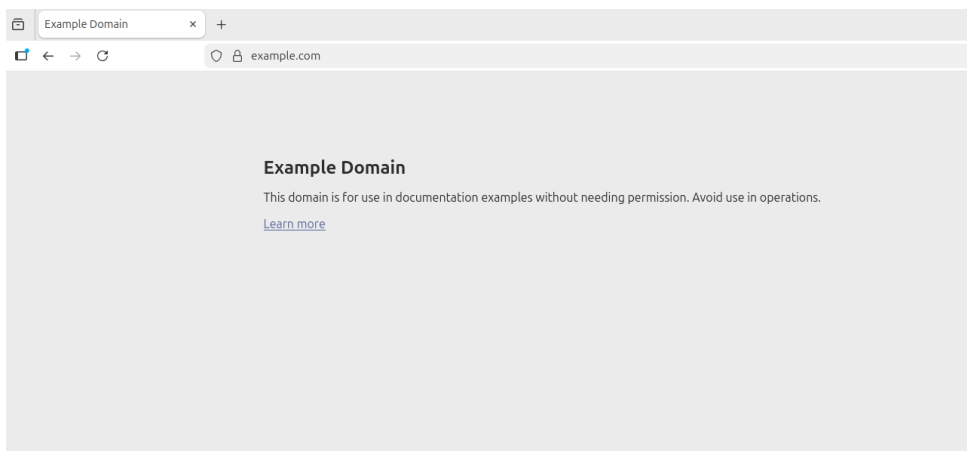
Command - The command used to capture data packets while visiting the website was



```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
irman-sharma@irman-sharma-VirtualBox:~/Desktop$ sudo tcpdump -i any -nn port 443
```

sudo tcpdump -i any -nn port 443. **sudo** is used to run a command with elevated privileges, **tcpdump** starts capturing packets, **-i any** is used to listen/capture traffic passing through all network interfaces. **-nn** is used so that the IP addresses or port numbers are not resolved to hostnames and service names respectively. **port 443** is a filter used to capture only HTTPS traffic.

Website -



The website visited on web browser Firefox was <https://example.com/>

TCP three-way handshake -

```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
[irman-sharma@irman-sharma-VirtualBox:~/Desktop$ sudo tcpdump -i any -nn port 443
[sudo] password for irman-sharma:
Sorry, try again.
[sudo] password for irman-sharma:
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
13:33:16.082483 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [S], seq 1855709805,
  win 64240, options [mss 1460,sackOK,TS val 3333018201 ecr 0,nop,wscale 9], length 0
13:33:16.084350 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [S], seq 635817906,
  win 64240, options [mss 1460,sackOK,TS val 3333018203 ecr 0,nop,wscale 9], length 0
13:33:16.100368 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [S.], seq 403464705,
  ack 1855709806, win 65535, options [mss 1460], length 0
13:33:16.100398 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [.], ack 1, win 6424
  0, length 0
13:33:16.100596 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [P.], seq 1:221, ack
  1, win 64240, length 220
13:33:16.100706 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [.], ack 221, win 65
  535, length 0
13:33:16.103452 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [S.], seq 403528705,
  ack 635817907, win 65535, options [mss 1460], length 0
13:33:16.103480 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [.], ack 1, win 6424
  0, length 0
13:33:16.103653 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [P.], seq 1:221, ack
  1, win 64240, length 220
13:33:16.103762 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [.], ack 221, win 65
  535, length 0
```

In these logs, the initial TCP three-way handshake is visible: The browser sends a **SYN** request with source IP 10.0.2.15 to the web server through temporary port 48950 over port 443 (HTTPS). The IP address of the web server is 34.160.144.191. The packet with **Flags [S]** sends the **SYN** request to the web server. In another packet with **Flags [S.]**, the web server acknowledges the request (**SYN-ACK**). The last packet of the TCP handshake contains a final acknowledgement (**ACK**). When this packet with **Flags [.]** is sent to the server, the TCP connection is established.

Multiple TCP connections established simultaneously-

It's worth noting that the browser has sent another **SYN** packet to the server through another temporary port 48962. Firefox attempted two separate TCP connections through two different source ports. The request for the second connection was sent before the first connection was established; the connections started simultaneously.

Modern web browsers often establish multiple simultaneous TCP connections to retrieve webpage resources in parallel, improving page load performance.

Initial encrypted data transfer -

The browser sends the initial encrypted data to the server over the newly established connections. This data is probably related to the TLS handshake. The packets containing Flags [P.] represent this data transfer, and the web server acknowledges it with Flag [.]

After the initial encrypted data transfer, these logs follow:

```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
13:33:16.192875 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [..], seq 1:1441, ack 221, win 65535, length 1440
13:33:16.192877 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [..], seq 1441:2881, ack 221, win 65535, length 1440
13:33:16.192928 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [..], ack 1441, win 65535, length 0
13:33:16.192943 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [..], ack 2881, win 65535, length 0
13:33:16.193087 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [..], seq 2881:4321, ack 221, win 65535, length 1440
13:33:16.193088 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [P.], seq 4321:4508, ack 221, win 65535, length 187
13:33:16.193092 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [..], ack 4321, win 65535, length 0
13:33:16.193098 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [..], ack 4508, win 65535, length 0
13:33:16.193995 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [P.], seq 221:314, ack 4508, win 65535, length 93
13:33:16.194127 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [..], ack 314, win 65535, length 0
13:33:16.201530 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [..], seq 1:1441, ack 221, win 65535, length 1440
13:33:16.201531 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [..], seq 1441:2881, ack 221, win 65535, length 1440
13:33:16.201547 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [..], ack 1441, win 65535, length 0
13:33:16.201558 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [..], ack 2881, win 65535, length 0
```

```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
13:33:16.193092 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [..], ack 4321, win 65535, length 0
13:33:16.193098 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [..], ack 4508, win 65535, length 0
13:33:16.193995 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [P.], seq 221:314, ack 4508, win 65535, length 93
13:33:16.194127 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [..], ack 314, win 65535, length 0
13:33:16.201530 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [..], seq 1:1441, ack 221, win 65535, length 1440
13:33:16.201531 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [..], seq 1441:2881, ack 221, win 65535, length 1440
13:33:16.201547 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [..], ack 1441, win 65535, length 0
13:33:16.201558 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [..], ack 2881, win 65535, length 0
13:33:16.201702 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [..], seq 2881:4321, ack 221, win 65535, length 1440
13:33:16.201702 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 4321:4508, ack 221, win 65535, length 187
13:33:16.201707 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [..], ack 4321, win 65535, length 0
13:33:16.201713 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [..], ack 4508, win 65535, length 0
13:33:16.207175 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [P.], seq 221:314, ack 4508, win 65535, length 93
13:33:16.207368 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [..], ack 314, win 65535, length 0
```

In these logs, it's visible that the server is sending encrypted data packets to the client in chunks. Some of it still might be TLS handshake data. Once the TLS connection is established, the HTTP GET request is sent to the server and the application data is returned. What's inside the packets cannot be told for sure because they are encrypted.

In the following logs, encrypted data transfer between the web server and both the temporary ports continues-

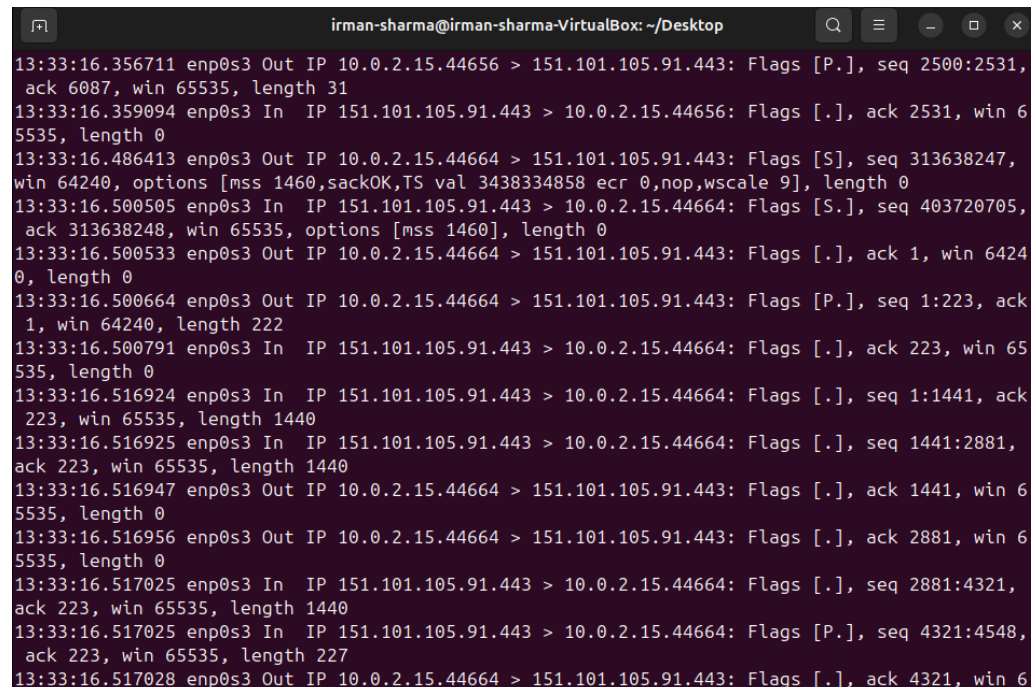
```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
5535, length 0
13:33:16.201713 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [P.], seq 221:314, ack 4508, win 65535, length 0
13:33:16.207175 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [P.], seq 221:314, ack 4508, win 65535, length 93
13:33:16.207368 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 314:413, ack 4888, win 65535, length 0
13:33:16.208662 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [P.], seq 4508:4888, ack 314, win 65535, length 380
13:33:16.208956 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [P.], seq 314:413, ack 4888, win 65535, length 99
13:33:16.209032 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [P.], seq 413:451, ack 4888, win 65535, length 38
13:33:16.209074 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [P.], seq 4888:4926, ack 451, win 65535, length 0
13:33:16.224140 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48950: Flags [P.], seq 4888:4926, ack 451, win 65535, length 38
13:33:16.227387 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 4508:4888, ack 314, win 65535, length 380
13:33:16.230325 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [P.], seq 314:430, ack 4888, win 65535, length 116
13:33:16.230428 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [FP.], seq 430:461, ack 4888, win 65535, length 31
13:33:16.230541 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 4888:4926, ack 462, win 65535, length 0
13:33:16.249582 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 4888:4926, ack 462, win 65535, length 38
```

```
ck 4888, win 65535, length 116
13:33:16.230428 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [FP.], seq 430:461, ack 4888, win 65535, length 31
13:33:16.230541 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 4888:4926, ack 462, win 65535, length 0
13:33:16.249582 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [P.], seq 4888:4926, ack 462, win 65535, length 38
13:33:16.249583 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [F.], seq 4926, ack 462, win 65535, length 0
13:33:16.249603 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [R], seq 635818368, win 0, length 0
13:33:16.249619 enp0s3 Out IP 10.0.2.15.48962 > 34.160.144.191.443: Flags [R], seq 635818368, win 0, length 0
13:33:16.249745 enp0s3 In IP 34.160.144.191.443 > 10.0.2.15.48962: Flags [R.], seq 3891438591, ack 462, win 0, length 0
13:33:16.264768 enp0s3 Out IP 10.0.2.15.48950 > 34.160.144.191.443: Flags [P.], seq 4888:4926, ack 462, win 65535, length 0
```

Encrypted data exchange between the server and the browser continues. It is visible in a particular packet how one of the temporary ports finishes the connection on its end gracefully (Flags [FP .]). The server also ends the connection established with this port (Flags [F .]). After the connection is ended gracefully, it is terminated abruptly (Flags [R .])

Multiple TCP connections -

```
5535, length 0
13:33:16.287971 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [S], seq 1707380310,
win 64240, options [mss 1460,sackOK,TS val 3438334660 ecr 0,nop,wscale 9], length 0
13:33:16.306447 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [S.], seq 403656705,
ack 1707380311, win 65535, options [mss 1460], length 0
13:33:16.306475 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [.], ack 1, win 6424
0, length 0
13:33:16.307025 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [P.], seq 1:1901, ac
k 1, win 64240, length 1900
13:33:16.307174 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [.], ack 1461, win 6
5535, length 0
13:33:16.307175 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [.], ack 1901, win 6
5535, length 0
13:33:16.327833 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [.], seq 1:1441, ack
1901, win 65535, length 1440
13:33:16.327834 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [.], seq 1441:2881,
ack 1901, win 65535, length 1440
13:33:16.327861 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [.], ack 1441, win 6
5535, length 0
13:33:16.327875 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [.], ack 2881, win 6
5535, length 0
```



```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
13:33:16.356711 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [P.], seq 2500:2531,
ack 6087, win 65535, length 31
13:33:16.359094 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [.], ack 2531, win 6
5535, length 0
13:33:16.486413 enp0s3 Out IP 10.0.2.15.44664 > 151.101.105.91.443: Flags [S], seq 313638247,
win 64240, options [mss 1460,sackOK,TS val 3438334858 ecr 0,nop,wscale 9], length 0
13:33:16.500505 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [S.], seq 403720705,
ack 313638248, win 65535, options [mss 1460], length 0
13:33:16.500533 enp0s3 Out IP 10.0.2.15.44664 > 151.101.105.91.443: Flags [.], ack 1, win 6424
0, length 0
13:33:16.500664 enp0s3 Out IP 10.0.2.15.44664 > 151.101.105.91.443: Flags [P.], seq 1:223, ack
1, win 64240, length 222
13:33:16.500791 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [.], ack 223, win 65
535, length 0
13:33:16.516924 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [.], seq 1:1441, ack
223, win 65535, length 1440
13:33:16.516925 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [.], seq 1441:2881,
ack 223, win 65535, length 1440
13:33:16.516947 enp0s3 Out IP 10.0.2.15.44664 > 151.101.105.91.443: Flags [.], ack 1441, win 6
5535, length 0
13:33:16.516956 enp0s3 Out IP 10.0.2.15.44664 > 151.101.105.91.443: Flags [.], ack 2881, win 6
5535, length 0
13:33:16.517025 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [.], seq 2881:4321,
ack 223, win 65535, length 1440
13:33:16.517025 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [P.], seq 4321:4548,
ack 223, win 65535, length 227
13:33:16.517028 enp0s3 Out IP 10.0.2.15.44664 > 151.101.105.91.443: Flags [.], ack 4321, win 6
```

In these logs, it is visible that more TCP three-way handshakes are established. These connections are normally established by modern web browsers as new communication requirements arise during a browsing session, including requests for additional webpage resources and background network activity.

Sudden burst of SYN packets -

```
535, length 0
13:33:16.541638 enp0s3 Out IP 10.0.2.15.39646 > 34.111.211.209.443: Flags [S], seq 1316889164,
win 64240, options [mss 1460,sackOK,TS val 3440891013 ecr 0,nop,wscale 9], length 0
13:33:16.541688 enp0s3 Out IP 10.0.2.15.39650 > 34.111.211.209.443: Flags [S], seq 1185201702,
win 64240, options [mss 1460,sackOK,TS val 3440891013 ecr 0,nop,wscale 9], length 0
13:33:16.542791 enp0s3 Out IP 10.0.2.15.39658 > 34.111.211.209.443: Flags [S], seq 98597740, w
in 64240, options [mss 1460,sackOK,TS val 3440891015 ecr 0,nop,wscale 9], length 0
13:33:16.542903 enp0s3 Out IP 10.0.2.15.39668 > 34.111.211.209.443: Flags [S], seq 1918128128,
win 64240, options [mss 1460,sackOK,TS val 3440891015 ecr 0,nop,wscale 9], length 0
13:33:16.542947 enp0s3 Out IP 10.0.2.15.39670 > 34.111.211.209.443: Flags [S], seq 887939185,
win 64240, options [mss 1460,sackOK,TS val 3440891015 ecr 0,nop,wscale 9], length 0
13:33:16.542976 enp0s3 Out IP 10.0.2.15.39676 > 34.111.211.209.443: Flags [S], seq 3756644268,
win 64240, options [mss 1460,sackOK,TS val 3440891015 ecr 0,nop,wscale 9], length 0
13:33:16.543190 enp0s3 Out IP 10.0.2.15.41996 > 34.120.237.76.443: Flags [S], seq 2719670394,
win 64240, options [mss 1460,sackOK,TS val 1492506401 ecr 0,nop,wscale 9], length 0
13:33:16.543219 enp0s3 Out IP 10.0.2.15.42006 > 34.120.237.76.443: Flags [S], seq 3297647230,
win 64240, options [mss 1460,sackOK,TS val 1492506401 ecr 0,nop,wscale 9], length 0
13:33:16.543249 enp0s3 Out IP 10.0.2.15.42012 > 34.120.237.76.443: Flags [S], seq 1501465821,
win 64240, options [mss 1460,sackOK,TS val 1492506401 ecr 0,nop,wscale 9], length 0
13:33:16.543284 enp0s3 Out IP 10.0.2.15.42014 > 34.120.237.76.443: Flags [S], seq 1445316148,
win 64240, options [mss 1460,sackOK,TS val 1492506401 ecr 0,nop,wscale 9], length 0
13:33:16.543381 enp0s3 Out IP 10.0.2.15.42028 > 34.120.237.76.443: Flags [S], seq 73061331, wi
n 64240, options [mss 1460,sackOK,TS val 1492506401 ecr 0,nop,wscale 9], length 0
13:33:16.543429 enp0s3 Out IP 10.0.2.15.42032 > 34.120.237.76.443: Flags [S], seq 3351672817,
win 64240, options [mss 1460,sackOK,TS val 1492506401 ecr 0,nop,wscale 9], length 0
13:33:16.548010 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44664: Flags [P.], seq 4806:4878,
ack 415, win 65535, length 72
```

In these logs, multiple **SYN** packets sent to multiple IP addresses from multiple temporary ports are visible. Firefox is opening many connections almost simultaneously. Modern browsers might do this for parallel resource loading, different backend or CDN servers, browser backend services or alternative connection attempts. However, there is one important point to be noted: all port 443 traffic was captured from the VM, not only traffic associated with [example.com](#). Hence not every IP address in these logs is necessarily part of the example.com page.

UDP Traffic on Port 443 -


```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
13:33:18.344121 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 1252
13:33:18.344124 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 1252
13:33:18.360276 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 43
13:33:18.363503 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.363604 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.363605 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.363672 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.363673 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.363724 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1058
13:33:18.373110 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 1252
13:33:18.376317 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 131
13:33:18.376348 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 78
13:33:18.376733 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 399
13:33:18.391289 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 618
13:33:18.391436 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 124
13:33:18.391437 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 26
13:33:18.391802 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 31
13:33:18.396269 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1248
13:33:18.396324 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.396384 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.396452 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.396630 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 36
13:33:18.397827 enp0s3 Out IP 10.0.2.15.53780 > 142.251.155.119.443: UDP, length 188
13:33:18.401146 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.401201 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 1252
13:33:18.401241 enp0s3 In IP 142.250.182.110.443 > 10.0.2.15.36849: UDP, length 151
13:33:18.401362 enp0s3 Out IP 10.0.2.15.36849 > 142.250.182.110.443: UDP, length 31
```

In these logs, packets using “UDP” over port 443 are visible. This is likely QUIC/HTTP/3 traffic, used by many modern websites. HTTP/3 commonly uses the QUIC transport protocol over UDP port 443 while providing secure, encrypted web communication similar to HTTPS over TCP.

Overall Pattern

```
irman-sharma@irman-sharma-VirtualBox: ~/Desktop
13:33:16.996912 enp0s3 In IP 34.120.237.76.443 > 10.0.2.15.42006: Flags [..], ack 6189, win 65535, length 0
13:33:17.002469 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [S..], seq 404616705, ack 515821428, win 65535, options [mss 1460], length 0
13:33:17.002502 enp0s3 Out IP 10.0.2.15.53838 > 151.101.1.91.443: Flags [..], ack 1, win 64240, length 0
13:33:17.004387 enp0s3 Out IP 10.0.2.15.53838 > 151.101.1.91.443: Flags [P..], seq 1:1905, ack 1, win 64240, length 1904
13:33:17.004520 enp0s3 Out IP 10.0.2.15.42006 > 34.120.237.76.443: Flags [P..], seq 6189:6431, ack 6501, win 65535, length 242
13:33:17.004632 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [..], ack 1461, win 65535, length 0
13:33:17.004633 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [..], ack 1905, win 65535, length 0
13:33:17.004633 enp0s3 In IP 34.120.237.76.443 > 10.0.2.15.42006: Flags [..], ack 6431, win 65535, length 0
13:33:17.007786 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [S..], seq 404680705, ack 1995843525, win 65535, options [mss 1460], length 0
13:33:17.007924 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [..], ack 1, win 64240, length 0
13:33:17.008716 enp0s3 In IP 34.120.237.76.443 > 10.0.2.15.42006: Flags [P..], seq 6501:6616, ack 6431, win 65535, length 115
13:33:17.008809 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [P..], seq 1:1905, ack 1, win 64240, length 1904
13:33:17.009032 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [..], ack 1461, win 65535, length 0
13:33:17.009032 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [..], ack 1905, win 65535, length 0
```



```

irman-sharma@irman-sharma-VirtualBox: ~/Desktop
13:33:17.074653 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [..], ack 6022, win 655
35, length 0
13:33:17.076446 enp0s3 Out IP 10.0.2.15.53838 > 151.101.1.91.443: Flags [P.], seq 1905:1969, a
ck 6022, win 65535, length 64
13:33:17.076639 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [..], ack 1969, win 655
35, length 0
13:33:17.076688 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [P.], seq 1905:1969, a
ck 6022, win 65535, length 64
13:33:17.076826 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [..], ack 1969, win 655
35, length 0
13:33:17.077253 enp0s3 Out IP 10.0.2.15.53838 > 151.101.1.91.443: Flags [P.], seq 1969:2061, a
ck 6022, win 65535, length 92
13:33:17.077292 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [P.], seq 1969:2078, a
ck 6022, win 65535, length 109
13:33:17.077420 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [..], ack 2061, win 655
35, length 0
13:33:17.077506 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [..], ack 2078, win 655
35, length 0
13:33:17.077626 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [P.], seq 2078:2102, a
ck 6022, win 65535, length 24
13:33:17.077649 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [F.], seq 2102, ack 60
22, win 65535, length 0
13:33:17.077774 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [..], ack 2102, win 655
35, length 0
13:33:17.077775 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [..], ack 2103, win 655
35, length 0
13:33:17.138853 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [P.], seq 6022:6087, a

```

```

2103, win 65535, length 24
13:33:17.138963 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [R], seq 1995845627, w
in 0, length 0
13:33:17.139049 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [F.], seq 6046, ack 21
03, win 65535, length 0
13:33:17.139054 enp0s3 Out IP 10.0.2.15.53824 > 151.101.1.91.443: Flags [R], seq 1995845627, w
in 0, length 0
13:33:17.139245 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53824: Flags [R.], seq 3890286591,
ack 2103, win 0, length 0
13:33:17.149909 enp0s3 Out IP 10.0.2.15.53838 > 151.101.1.91.443: Flags [P.], seq 2061:2092, a
ck 6087, win 65535, length 31
13:33:17.153179 enp0s3 In IP 151.101.1.91.443 > 10.0.2.15.53838: Flags [..], ack 2092, win 655
35, length 0
13:33:17.253158 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [P.], seq 14049:1424
ack 3756, win 65535, length 192
13:33:17.257393 enp0s3 In IP 151.101.105.91.443 > 10.0.2.15.44656: Flags [P.], seq 14241:1442
ack 3756, win 65535, length 185
13:33:17.257481 enp0s3 Out IP 10.0.2.15.44656 > 151.101.105.91.443: Flags [..], ack 14426, win

```

In the entire log output of the captured packets, certain patterns are visible throughout -

- Browser starts several connections
- TCP handshakes occur
- TLS negotiation data is exchanged
- Encrypted requests and responses follow
- Data arrives in numbered chunks
- ACKs confirm receipt
- Connections close gracefully with FIN

- Then the connections are abruptly ended with RST
- Some traffic uses UDP/QUIC instead of TCP

Key Takeaways

- Successfully identified multiple TCP three-way handshakes
- Observed multiple simultaneous TCP connections
- Observed encrypted TLS traffic over HTTPS
- Observed data being delivered in numbered chunks, and TCP acknowledgments
- Observed graceful connection termination using FIN packets
- Observed abrupt connection termination using RST packets

Conclusion

This home lab demonstrated how a secure HTTPS connection is established and maintained. Using tcpdump, it was possible to identify the TCP three-way handshake, encrypted data exchange, multiple concurrent TCP connections, acknowledgments, and connection termination. Although the encrypted application data could not be inspected, the packet metadata provided valuable insight into the communication process.